

Trigonometry

You are expected to have the following facts memorized.

$$1. \sin x = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$2. \cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$3. \tan x = \frac{\text{opposite}}{\text{adjacent}}$$

$$4. \tan x = \frac{\sin x}{\cos x} = \frac{1}{\cot x}$$

$$5. \sec x = \frac{1}{\cos x}$$

$$6. \csc x = \frac{1}{\sin x}$$

$$7. \sin^2 x + \cos^2 x = 1$$

$$8. \tan^2 x + 1 = \sec^2 x$$

$$9. 1 + \cot^2 x = \csc^2 x$$

$$10. \sin(2x) = 2 \sin x \cos x$$

$$11. \cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$12. \sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$13. \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$14. \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$15. e^{ix} = \cos x + i \sin x$$

$$16. \cosh x = \frac{e^x + e^{-x}}{2}$$

$$17. \sinh x = \frac{e^x - e^{-x}}{2}$$

$$18. e^x = \cosh x + \sinh x$$

You should also know the the graphs of the trig functions and hyperbolic trig functions, as well as the values of the trig functions for the angles $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{\pi}{4},$ and $\frac{\pi}{3}$.

Derivatives

You are expected to have the following facts memorized.

$$1. \frac{d}{dx} x^n = nx^{n-1}$$

$$2. \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$3. \frac{d}{dx} \ln x = \frac{1}{x}$$

$$4. \frac{d}{dx} \log_b x = \frac{1}{x} \frac{1}{\ln b}$$

$$5. \frac{d}{dx} e^x = e^x$$

$$6. \frac{d}{dx} a^x = a^x \ln a$$

$$7. \frac{d}{dx} \sin x = \cos x,$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$8. \frac{d}{dx} \tan x = \sec^2 x,$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$9. \frac{d}{dx} \sec x = \sec x \tan x,$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

$$10. \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$11. \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2},$$

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$12. \frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}},$$

$$\frac{d}{dx} \csc^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$$

$$13. \frac{d}{dx} \sinh x = \cosh x,$$

$$\frac{d}{dx} \cosh x = \sinh x$$

Integrals

You are expected to have the following facts memorized.

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

$$2. \int \frac{1}{x} dx = \ln |x| + C$$

$$3. \int e^x dx = e^x + C$$

$$4. \int a^x dx = a^x \frac{1}{\ln a} + C$$

$$5. \int \sin x dx = -\cos x + C,$$

$$\int \cos x dx = \sin x + C$$

$$6. \int \tan x dx = \ln |\sec x| + C,$$

$$\int \cot x dx = -\ln |\csc x| + C$$

$$7. \int \sec x dx = \ln |\sec x + \tan x| + C,$$

$$\int \csc x dx = -\ln |\csc x + \cot x| + C$$

$$8. \int \sec^2 x dx = \tan x + C,$$

$$\int \csc^2 x dx = -\cot x + C$$

$$9. \int \sec x \tan x dx = \sec x + C,$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$10. \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$11. \int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$12. \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$13. \int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$$

$$14. \int \sinh x dx = \cosh x + C,$$

$$\int \cosh x dx = \sinh x + C$$